

Hoja de Trabajo 4

Ejercicio 1 longitud de arco [0, 2]

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$y = \frac{1}{2}(e^x + e^{-x})$$

$$y' = \frac{1}{2}(e^x - e^{-x}) = \left[\frac{1}{2}(e^x - e^{-x}) \right]^2$$

$$(y')^2 = \frac{1}{4}(e^{2x} - 2(e^x \cdot e^{-x}) + e^{-2x})$$
$$= \frac{1}{4}(e^{2x} - 2 + e^{-2x})$$

$$L = \int_0^2 \sqrt{1 + \left(\frac{1}{4}(e^{2x} - 2 + e^{-2x}) \right)} dx$$

$$L = \int_0^2 \sqrt{\frac{4 + e^{2x} - 2 + e^{-2x}}{4}} dx$$

$$L = \frac{1}{2} \int_0^2 \sqrt{4 + e^{2x} - 2 + e^{-2x}} dx =$$

$$= \frac{1}{2} \int_0^2 \sqrt{e^{2x} + e^{-2x} + 2} dx = \sqrt{(e^x + e^{-x})^2}$$

$$= \frac{1}{2} \int_0^2 (e^x + e^{-x}) dx = \left(e^x + (-e^{-x}) \right) \Big|_0^2 =$$

$$= \frac{1}{2}(e^x - e^{-x}) \Big|_0^2 = 3.63$$

Ejercicio 2

area superficial

$$y = 9 - x^2$$

de y [0, 3]

$$y' = -2x =$$

$$(y')^2 = 4x^2$$

$$S = \int_0^3 2\pi x \sqrt{1+4x^2} dx$$

$$S = 2\pi \int_0^3 x \sqrt{1+4x^2} dx$$

$$S = 2\pi \int_0^3 x \sqrt{m} \frac{dm}{8x}$$

$$m = 1+4x^2$$

$$dm = 8x dx$$

$$S = \frac{\pi}{4} \int_0^3 \sqrt{m} dm = \frac{\pi}{4} \int_0^3 m^{1/2} dm$$

$$S = \frac{\pi}{4} \left(\frac{2}{3} m^{3/2} \right) = \frac{\pi}{4} \left(\frac{2}{3} (1+4x^2)^{3/2} \right) \Big|_0^3$$

$$S = 177.8422995 - \frac{\pi}{6} = \underline{177.32 \text{ u}^2}$$

Exercício 3

$$X - Y = 2, \quad X = Y^2$$

$$X = 2 + Y$$

$$\frac{dY}{dX} = 2 + Y$$

$$Y^2 - Y - 2 = 0$$

$$Y_1 = 2, \quad X_1 = 4$$

$$Y_2 = -1, \quad X_2 = 1$$

$$A = \int_{-1}^2 Y^2 - (2 + Y) dY$$

$$A = \int_{-1}^2 Y^2 - 2 - Y dY$$

$$A = \left[\frac{1}{3} Y^3 - 2Y - \frac{1}{2} Y^2 \right]_{-1}^2$$

$$A = -\frac{10}{3} - \frac{7}{6} = -\frac{9}{2}$$

$$\bar{X} = \frac{1}{2(-\frac{9}{2})} \int_{-1}^2 \{ (Y^2)^2 - (2 + Y)^2 \} dY$$

$$\bar{X} = -\frac{1}{9} \int_{-1}^2 Y^4 - (4 + 4Y + Y^2) dY = -\frac{1}{9} \int_{-1}^2 Y^4 - 4 - 4Y - Y^2 dY$$

$$\bar{X} = -\frac{1}{9} \left(\frac{1}{5} Y^5 - 4Y - 2Y^2 - \frac{1}{3} Y^3 \right) \Big|_{-1}^2 = \frac{184}{135} - \frac{32}{135} = \frac{8}{5} = 1.6$$

$$\bar{Y} = \frac{1}{-\frac{9}{2}} \int_{-1}^2 Y [Y^2 - (2 + Y)] dY$$

$$\bar{Y} = -\frac{2}{9} \int_{-1}^2 Y(Y^2 - 2 - Y) dY = -\frac{2}{9} \int_{-1}^2 Y^3 - 2Y - Y^2 dY$$

$$\bar{Y} = -\frac{2}{9} \left(\frac{1}{4} Y^4 - Y^2 - \frac{1}{3} Y^3 \right) \Big|_{-1}^2 = \frac{16}{27} - \frac{5}{54} = \frac{1}{2} = 0.5$$

$$\rho = 62.4 \text{ lb/ft}^3$$

$$m = 9800 \text{ N/m}$$

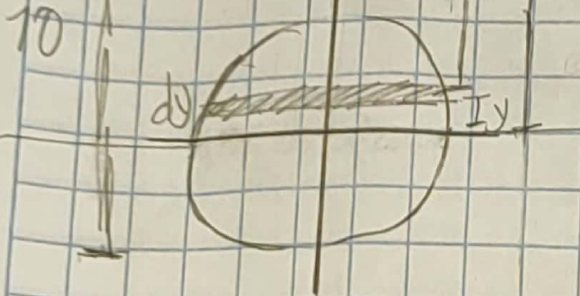
Ejercicio 4 fuerza

$$F = \rho g d A$$

$$\rho g = 62.4$$

$$d = 7 - y$$

$$A = 2x dy = 2\sqrt{9-y^2} dy$$



$$x^2 + y^2 = 9$$

$$x = \sqrt{9-y^2}$$

$$F = \int_{-3}^3 62.4 (7-y) 2(\sqrt{9-y^2}) dy$$

$$F = 124.8 \int_{-3}^3 (7-y) \sqrt{9-y^2} dy$$

$$F = 124.8 \int_{-3}^3 7\sqrt{9-y^2} - y\sqrt{9-y^2} dy$$

$$F = 124.8 \left[7 \int_{-3}^3 \sqrt{9-y^2} dy - \int_{-3}^3 y\sqrt{9-y^2} dy \right]$$

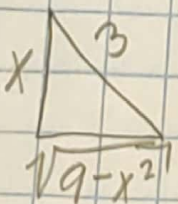
$$F = 873.6 \left[\int_{-\pi/2}^{\pi/2} \sqrt{9-(9\sin^2\theta)} 3\cos\theta d\theta - \int_{-\pi/2}^{\pi/2} \sqrt{1-\sin^2\theta} 3\cos\theta d\theta \right]$$

$$x = 3\sin\theta \quad dx = 3\cos\theta d\theta \quad = 9 \int_{-\pi/2}^{\pi/2} \frac{4\cos^2\theta}{2} d\theta$$

$$\frac{x}{3} = \sin\theta$$

$$x = -3, -\frac{\pi}{2}$$

$$x = 3, \frac{\pi}{2}$$



$$m = 2\theta$$

$$dm = 2d\theta$$

$$= \frac{9}{2} \int_{-\pi/2}^{\pi/2} 1 + \cos 2\theta d\theta = \cos m \frac{dm}{2}$$

$$= \frac{9}{2} \left(\theta + \frac{\sin 2\theta}{2} \right)$$

$$= \frac{9}{2} \left[\sin^{-1}\left(\frac{y}{3}\right) + 2\left(\frac{x}{3}\right)\left(\frac{\sqrt{9-x^2}}{3}\right) \right]$$

So ca To

$$= \frac{9828}{5} \pi - \left(-\frac{9828}{5} \pi \right) = \frac{19656}{5} \pi$$